



ashoka
UNIVERSITY

Physics Handbook

GUIDELINES FOR STUDENTS

October 2024 Update



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Department of Physics

Introduction

Physics is, simultaneously, a doorway to some of the most beautiful and profound phenomena in the universe, e.g. black holes, supernovae, Bose-Einstein condensates, superconductors; a driver of lifestyle-changing technology, e.g. engines, electricity, and transistors; and a powerful way of perceiving and analyzing problems that can be applied in various domains, both within and outside standard physics. The beauty and profundity of the phenomena studied by physicists offer romance and excite passion, and the utility of its discoveries and the power of its methods arouse interest. These methods can be very intricate and demanding: theoretical physics requires a skillful combination of physical and mathematical thinking, and experimental physics requires in addition the ability to turn tentative ideas into physical devices that can put those ideas to the test. As a result, the successful practice of physics demands rigour, flexibility, mechanical adroitness, persistence, and great imagination. The physicist's imagination is nourished not just by physics but also by other areas of human enquiry and thought, of the kind that an Ashoka undergraduate is expected to encounter.

With all of this in mind, the physics programme has been designed to: (i) allow students wishing to major in physics to discover real physics and make a wise choice, in the first year; (ii) provide a thorough training in fundamental physics, in the second year; and (iii) bring together everything learnt earlier and give students the option to pursue more advanced courses in physics or branch out into other areas, in the third year. The idea is to accompany those wishing to become professional physicists as they take the first steps in that direction and to introduce everyone who goes through the programme to the physicist's way of thinking.

Faculty

Permanent Faculty Members



Somak Raychaudhury (Vice-Chancellor)
Professor of Physics
Ph.D. University of Cambridge



Gautam Menon (Dean, Research)
Professor of Physics and Biology,
Ph.D. Indian Institute of Science, Bangalore



Bikram Phookun (Dean, Academic Affairs)
Professor of Physics,
Ph.D. University of Maryland



Dipankar Bhattacharya (Head of the Department)
Sunanda and Santimay Basu Professor of Astrophysics,
Ph.D. Indian Institute of Science, Bangalore



Somendra M. Bhattacharjee
Professor of Physics,
Ph.D. Carnegie Mellon University



Amin A. Nizami
Assistant Professor of Physics,
Ph.D. University of Cambridge



Pramoda Kumar
Assistant Professor of Physics,
Ph.D. CeNS, Bangalore



Susmita Saha
Assistant Professor of Physics,
Ph.D. S. N. Bose National Center for Basic Science, Kolkata



Suratna Das
Assistant Professor of Physics,
Ph.D. Physical Research Laboratory, Ahmedabad



Garima Mishra
Assistant Professor of Physics,
Ph.D. Banaras Hindu University, Varanasi

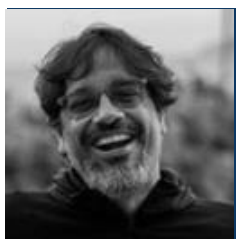
Visiting Faculty Members



Kandaswamy Subramanian
Visiting Professor of Astrophysics,
Ph.D. Tata Institute of Fundamental Research
(University of Bombay)



Sujan Sengupta
Visiting Professor of Physics,
Ph.D. Indian Institute of Astrophysics



Arjendu K. Pattanayak
Visiting Professor of Physics,
Ph.D. The University of Texas, Austin

Requirements for a Major/Minor/Concentration

☐ Physics Major

To pursue a major in Physics in 3 years, a student must complete **15 courses (60 credits)**:

2 Gateway Courses

11 Core Courses

2 Elective Courses

To pursue a major in Physics in 4 years, a student must complete **20 courses (80 credits)**:

2 Gateway Courses

11 Core Courses

7 Elective Courses

For 4-year Major with research, **12 research credits** are to be taken in addition to the above.

☐ Physics Minor

To pursue a minor in Physics, a student must complete **6 courses (24 credits)**:

2 Gateway Courses

2 Core Courses

2 Core/Elective Courses

☐ Physics Concentration

To pursue a concentration in Physics, a student must complete **4 courses (16 credits)**:

2 Gateway Courses

2 Core Courses

❑ Astronomy Minor

To pursue a minor in Astronomy, a student must complete **6 courses (24 credits)**:

2 Astronomy Core Courses

4 Astronomy Elective Courses

❑ Astronomy Concentration

To pursue a concentration in Astronomy, a student must complete **4 courses (16 credits)**:

1 Astronomy Core Course

3 Astronomy Core/Elective Courses

General Information on the Physics Courses

- Mathematical level — All theory courses are calculus-based. The level of mathematical sophistication will increase progressively. The mathematical physics courses in semesters 2 and 3 will be application-oriented rather than proof-oriented.
- Labs — All lab courses will involve extensive use of instruments to make observations. These experiments will in general illustrate ideas studied in the accompanying theory courses.
- The use of computers — All theory courses will include computational exercises, generally using the programming language Python. Labs will also require the use of computers.
- Duration — Theory: two lectures a week each lasting 1.5 hours. Lab: two 3-hour lab session per week. (In addition, there may be a one-hour tutorial per week for each theory course.)

Recommended Trajectories

B.Sc. Hons. In Physics (3 years)

YEAR 1		YEAR 2		YEAR 3	
MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING
FC	A Physics Primer	Classical Mechanics	Mathematical Physics 2	Quantum Mechanics	Physics of Matter
		Electricity & Magnetism in Light of Relativity	Thermal Physics	Statistical Mechanics	FC
FC	Lab 1: Introduction to Physics Through Experiments	Lab 2: Classical Mechanics and Electro-magnetism	Oscillations, Waves & Optics	Elective 1	FC
FC	FC	Mathematical Physics 1: Mathematical and Computational Toolkit	Lab 3: Optics, Oscillations & Thermodynamics	Other	Elective 2
FC	FC				
	Other	FC	Other	Other	Other

B.Sc. Hons. In Physics (4 years) with and without research

YEAR 1		YEAR 2		YEAR 3		YEAR 4	
MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING
FC	A Physics Primer	Classical Mechanics	Mathematical Physics 2	Quantum Mechanics	Physics of Matter	Elective 4 (4000 level)	Elective 6 (4000 level)
		Electricity & Magnetism in Light of Relativity	Thermal Physics	Statistical Mechanics	FC	Elective 5 (4000 level)	Elective 7 (4000 level)
FC	Lab 1: Introduction to Physics Through Experiments	Lab 2: Classical Mechanics and Electro-magnetism	Oscillations, Waves & Optics	Elective 1	FC	Minor	Minor
FC	Minor	Mathematical Physics 1: Mathematical and Computational Toolkit	Lab 3: Optics, Oscillations & Thermodynamics	Elective 2	Elective 3	Research or 4 other courses	
FC	FC						
	FC	Minor	FC	Minor	Minor		

Recommended Trajectories

Delayed start in B.Sc. Hons. In Physics (4 years) without research

YEAR 1		YEAR 2		YEAR 3		YEAR 4	
MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING
FC	Other	Other	A Physics Primer	Classical Mechanics	Mathematical Physics 2	Quantum Mechanics	Physics of Matter
FC	Other	Other	Lab 1: Introduction to Physics Through Experiments	Electricity & Magnetism in Light of Relativity	Thermal Physics	Statistical Mechanics	Physics/Astro Elective 4 (4000 level)
FC	FC			Lab 2: Classical Mechanics and Electro-magnetism	Oscillations, Waves & Optics	Physics/Astro Elective 2	Physics/Astro Elective 5 (4000 level)
FC	FC	Other	Other	Physics/Astro Elective 1	Lab 3: Optics, Oscillations & Thermodynamics	Physics/Astro Elective 3	Physics/Astro Elective 6 (4000 level)
	FC	Other	Other	Mathematical Physics 1: Mathematical and Computational Toolkit	Other	Other	Physics/Astro Elective 7 (4000 level)

Recommended Trajectories

B.Sc. Hons. In Physics (4 years) with Astronomy minor

YEAR 1		YEAR 2		YEAR 3		YEAR 4	
MON-SOON	SPRING	MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING
FC	A Physics Primer	Classical Mechanics	Mathematical Physics 2	Quantum Mechanics	Physics of Matter	Physics/ Astro Elective 4 (4000 level)	Physics/ Astro Elective 6 (4000 level)
		Electricity & Magnetism in Light of Relativity	Thermal Physics	Statistical Mechanics	FC		
FC	Lab 1: Introduction to Physics Through Experiments	Lab 2: Classical Mechanics and Electro-magnetism	Oscillations, Waves & Optics	Physics/ Astro Elective 1	FC	Physics/ Astro Elective 5 (4000 level)	Physics/ Astro Elective 7 (4000 level)
FC	Observing the Cosmos	Physics of the Universe	Lab 3: Optics, Oscillations & Thermodynamics	Physics/ Astro Elective 2	Physics/ Astro Elective 3	Other	Other
FC	FC					Research or 4 other courses	
	FC	Mathematical Physics 1: Mathematical and Computational Toolkit	FC	Other	Other		

Recommended Trajectories

B.Sc. Hons. In Physics (4 years) with Biology minor

YEAR 1		YEAR 2		YEAR 3		YEAR 4	
MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING
FC	A Physics Primer	Classical Mechanics	Mathematical Physics 2	Quantum Mechanics	Physics of Matter	Elective 4 (4000 level)	Elective 6 (4000 level)
FC	Lab 1: Introduction to Physics Through Experiments	Electricity & Magnetism in Light of Relativity	Thermal Physics	Statistical Mechanics	Elective 2	Elective 5 (4000 level)	Elective 7 (4000 level)
FC	FC	Lab 2: Classical Mechanics and Electromagnetism	Oscillations, Waves & Optics	Elective 1	Elective 3	Bio Elective 1	Bio Elective 2
FC	FC	Mathematical Physics 1: Mathematical and Computational Toolkit	Lab 3: Optics, Oscillations & Thermodynamics	FC	FC	Research or 4 other courses	
	Bio Core 1	Bio Core 2	FC	Bio Core 3	Bio Core 4		

Recommended Trajectories

B.Sc. Hons. In Physics (4 years) with Computer Science minor

YEAR 1		YEAR 2		YEAR 3		YEAR 4	
MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING	MONSOON	SPRING
FC	A Physics Primer	Classical Mechanics	Mathematical Physics 2	Quantum Mechanics	Physics of Matter	Elective 4 (4000 level)	Elective 6 (4000 level)
FC	Lab 1: Introduction to Physics Through Experiments	Electricity & Magnetism in Light of Relativity	Thermal Physics	Statistical Mechanics	Elective 2	Elective 5 (4000 level)	Elective 7 (4000 level)
FC	Introduction to Computer Programming	Lab 2: Classical Mechanics and Electro-magnetism	Oscillations, Waves & Optics	Elective 1	Elective 3	FC	FC
FC	CS Core 2	Mathematical Physics 1: Mathematical and Computational Toolkit	Lab 3: Optics, Oscillations & Thermodynamics	FC	FC	Research or 4 other courses	
	FC	CS Core 3	CS Core 4	CS Core 5/ CS Elective	CS Core 6/ CS Elective		

Physics ASP for UG 2024 and earlier batches

The fourth year ASP provides the opportunity for students to study physics at a level more advanced than the usual undergraduate level. Alternatively, students wishing to broaden their education further can use it to take a minor/concentration in any other subject or simply take whatever courses in any department they wish to.

Advanced Major

The advanced major is for students intending to pursue higher studies and research in physics. It would enable them to take some advanced physics courses as well as get an idea of research through the capstone project/thesis.

To pursue an advanced major in Physics, a student must complete **32 credits**:

4 4000 Level Physics Elective Courses

1 Capstone Thesis (8 Credits)

2 Other courses from any department

Capstone Project and Thesis

Interested students should talk to prospective advisors about possible projects in their 6th semester. After mutual agreement - the advisor agrees to guide the student and the student agrees to work on one of the projects suggested by the advisor - students should start with necessary background reading / preliminary work over the summer before the start of the fourth year.

At the end of the 7th semester the project would be assessed with a presentation and report - this will be evaluated by the physics faculty. Students without satisfactory progress at this stage may not continue for the thesis subsequently.

At the end of the 8th semester the thesis has to be submitted and defended before a panel which may also include some external members. It is mandatory to successfully complete the capstone project and thesis to be eligible for the advanced major.

Other options (Minor/Concentration)

Students not opting for the advanced major can take courses as per their interest in the fourth year. It is recommended that physics majors take two physics electives over the year. Those minoring in physics (or taking it as a concentration) may take physics courses accordingly to meet the necessary requirements.

Independent Study Module (ISM)

Students may opt for an ISM if interested. A successfully completed ISM would contribute **4 credits** to the student's GPA. Students who wish to take up an ISM can contact the Professor whose guidance they prefer seeking and can discuss the course load and hours to spend. Further instructions will be provided by the Professor.

TF/TA-ship Policy

The department doesn't currently offer TA-ship but there are possibilities of voluntarily working as TAs for the physics courses (for the 3rd and 4th year students). The Teaching Fellows are recruited after a departmental screening process. MLS students are assigned to be TAs as a part of their academic requirements.

Description of Physics Courses

Gateway Courses

Level 1000

PHY-1220 A Physics Primer

This course will provide a panorama of physics including both classical and modern topics. An appreciation that Physics involves understanding the dynamics of systems and the forces which govern these dynamics will be developed. Dimensional analysis and order of magnitude estimates will also be used, where possible, to build intuition for thinking physically, doing “back of envelope” calculations and making informed guesses. An overview of Newton’s Laws and classical physics, Kepler’s laws, Symmetry and conservation laws, energy released in nuclear detonation, electromagnetism, quantum physics, uncertainty principle, Bohr model H-atom, black body radiation, statistical physics, etc.

Pre-requisites:

The course will assume familiarity with calculus, algebra and trigonometry at an elementary level.

PHY1010 Lab 1: Introduction to Physics through Experiments

An introduction to physics through experiments. The idea behind this course is to help students, using simple experiments, to acquire the skills, protocols, and attitudes required to perform and design experiments and to analyze observations successfully, so that they become a part of his/her mind-set.

Pre-requisites:

None.

Core Courses

Level 1000

PHY1110 Mathematical Physics 1: Mathematical and Computational Toolkit

This course familiarize the student with a variety of mathematical techniques which every student of physics should be conversant with. Having taken the course, you should be comfortable with casting a wide variety of physics problems in mathematical language and being able to analyze and solve them subsequently. The course includes an introduction to programming with Python. Within the physics curriculum at Ashoka university, this course is an essential prerequisite for Classical Mechanics and Electricity and Magnetism in Light of Relativity offered in the third semester.

Pre-requisites: Students who have not done mathematics at the 12th-grade level are required to take Calculus I.

Level 2000

PHY2010 Lab 2: Classical Mechanics and Electromagnetism

The idea behind the first lab course (Introduction to Physics through Experiments) is to help the students, using simple and interesting experiments, to acquire the skills, protocols, and attitude required to perform and design experiments and to analyze observations successfully, so that students understand what experimental physics is all about. This Lab-2 course is a little bit more advanced compared to Lab-1. The experiments (list is given below) are chosen in such a way that they would help the students to understand the theoretical concepts taught in the classroom lectures. Lab-2 will have the following experiments broadly based on the principles of classical mechanics and electromagnetism.

1. Kater's Pendulum (to determine the acceleration due to gravity)
2. Air Track (to study conservation of momentum and energy)
3. Free Fall
4. Electromagnetic Damping
5. Equipotential Curves
6. Electronics Circuits (LCR Bridge)

Pre-requisites: Lab 1: Introduction with Physics through experiment

PHY2210 Classical Mechanics

The course provides a solid foundation in Classical Mechanics. We begin by formulating the primary concepts: the laws of motion and the dynamical equations as differential equations which need to be integrated to find the solutions of mechanical problems. We stress the importance of conservation laws not only in the solution of otherwise intractable dynamical problems but also their wider scope beyond Newtonian dynamics. As applications we deal with central force motion and the harmonic oscillator. Besides their historical importance, these are rich and instructive examples illustrating the basic concepts.

The second part of the course deals with Analytical Mechanics: We make the transition from Newtonian to Lagrangian dynamics by introducing the notion of generalised co-ordinates. The variational principle is then introduced as a unified way of dealing with diverse physical problems. Lagrange's equations are re-derived and the transition to the Hamiltonian form of dynamics is made thereby introducing the concept of phase space - a notion of special importance in statistical mechanics. Subsequently, the Poisson bracket formulation would be discussed. Throughout we stress the importance of, and the connections between, symmetry principles and conservation laws.

This course would be a steppingstone for later courses in quantum and statistical mechanics. It should also enable you to later study more advanced topics in classical physics like fluid mechanics, non-linear dynamical systems and chaos.

Pre-requisites: Students who have not done mathematics at the 12th-grade level are required to take Calculus I.

PHY2310 Electricity & Magnetism in Light of Relativity

The beautiful theory of electricity and magnetism is, with classical mechanics, the heart of classical physics. What is often not appreciated at the undergraduate level is that electricity and magnetism are related in a way that reveals the structure of space-time. In this course relativity is used from the beginning to relate electric and magnetic fields, so that their unity, as components of the electromagnetic field are revealed and used in the study of Maxwell's Equations.

Pre-requisites: Mathematical Physics I (Mathematical and Computational Toolkit).

PHY2110 Mathematical Physics 2

This mathematical physics course aims to be an introduction to differential equations with a view towards physical applications (particularly to later physics courses). The course is in two parts. Part I covers first order ODEs, non-linear systems and dynamical systems and in Part II we learn second order ODEs, special functions, Fourier analysis and PDEs.

Pre-requisites:

Multivariable calculus and partial differentiation besides the differential and integral calculus covered at the senior secondary school level. Some familiarity with basic linear algebra - eigenvalues and eigenvectors.

PHY2610 Thermal Physics

This course provides an introduction to thermodynamics, otherwise called “Thermal Physics”. Of this field, Einstein said, “ It is the only physical theory of universal content, which I am convinced, that within the framework of applicability of its basic concepts will never be overthrown.”

Given this unambiguous indication as to its importance, also as a foundation for other fields, this course develops the ideas and intuition appropriate to think about thermodynamics and make calculations in thermodynamics.

The approach of this course is first grounded in intuition regarding the need for conservation laws, such as of energy, expanded to include internal modes that are not accessible at the macro level. The difference between microscopic and macroscopic degrees of freedom is emphasized, and the problem of thermodynamics presented at understanding how behaviour in equilibrium at the macroscopic scale requires that we think of the impact of the microscopic degrees of freedom that must enter the description. We discuss the idea of entropy in detail, taking a postulational approach to thermodynamics and then show how all of what we take for granted in thermal equilibrium can be derived from a small number of basic principles.

Topics that this course cover include: the history of thermodynamic thought, fundamental relations in the energetic and entropic representations, the first, second and third law of thermodynamics, the Gibbs and Euler relations, Legendre transformations and free energies, Maxwell relations, Carnot cycles and their interpretation, useful work and the relation to statistical physics.

The emphasis are on (a) the development of the right intuition in its historical context (b) the codification of this intuition in terms of the laws of thermodynamics and how to understand them and (c) actual calculations of relevance to real life

PHY2020 Lab 3: Optics, Oscillations & Thermodynamics

Designed to accompany the theory courses - Thermal Physics and Oscillations, Waves & Optics, this laboratory course explores their experimental foundations. Students are introduced to the properties of light as a wave: interference, diffraction, and polarisation, through conceptually rich experiments like the Michelson Interferometer. In addition, mechanical systems which exhibit oscillatory behaviour, like coupled pendula and capillary waves, are also studied. Students are introduced to ideas of thermometry and calibration. The experiments list is given below

1. Coupled Pendulum
2. Interferometry and Polarization
3. Diffraction
4. Acoustics
5. Thermometry
6. Wien's displacement law and Stefan Boltzmann law

Pre-requisites: Lab 1: Introduction with Physics through experiment

PHY2410 Oscillations, Waves & Optics

Oscillatory phenomena appear in all areas of physics, both classical and quantum (to the point where a famous textbook begins by saying that the domain of physics is all phenomena that can be reduced to coupled oscillators). The methods used to study oscillatory motion are powerful and wide-ranging in their utility.

Pre-requisites: Classical Mechanics and Electricity & Magnetism in Light of Relativity

Level 3000

PHY3510 Quantum Mechanics

This course is an introduction to Quantum Mechanics (QM). Starting with a historical introduction and motivation, it goes on to introduce probability amplitudes as the fundamental physical basis for QM. Using the Dirac bra-ket notation, the fundamental postulates are then introduced, and their consequences worked out. The basic physical concepts are illustrated through spin- $\frac{1}{2}$ systems and one-dimensional wave mechanics. Symmetries and Conservation laws in QM are discussed, an introduction to the basic theory of angular momentum and spin is given, and the course culminates with the first major triumph of QM - a quantum mechanical understanding of the hydrogen atom.

Pre-requisites: Classical Mechanics and Electricity & Magnetism in Light of Relativity

PHY3710 Physics of Matter

In real physical systems the various areas of fundamental physics that are studied separately in semesters 3-5 are usually required all at once. The purpose of this course is to show how fundamental physics can be used to study a number of interesting phenomena. Recommended course content: Elastic properties of solids. Crystals and band structure. Semiconductors, insulators, and conductors. Electronic devices. Superconductivity. Glasses and amorphous materials. Liquids. Colloids. Polymers. Bose-Einstein condensation. The Chandrasekhar mass limit: white-dwarf and neutron stars.

Pre-requisites: All mandatory courses in the physics-major sequence.

PHY3610 Statistical Mechanics

Both Statistical Mechanics and Thermodynamics are ways of studying systems with large numbers of particles — of the order of the Avogadro number. When the number of particle is so large it becomes possible to describe systems in equilibrium in terms not just of quantities inherited from mechanics — particle-number, volume, and energy, etc — but also ones that arise from statistical regularities and are purely thermodynamic — entropy, temperature, and chemical potential, etc. In statistical mechanics we begin with an explicit acknowledgment of the particle nature of many-body systems, whereas in thermodynamics we ignore the particle nature and begin with relationships between thermodynamic quantities. They are thus complementary, and each allows a deeper understanding of the other.

In this course we use statistical mechanics to study simple classical and quantum systems, with and without interactions. We begin by understanding how the thermodynamic description of a simple system like an ideal gas in equilibrium can be found from statistical mechanics, and go on to more complex systems, both classical and quantum, that would be difficult to grasp using thermodynamics alone. We learn to describe fluctuations about equilibrium, which tell us about the stability of equilibrium states and are also related to equilibrium properties like specific heat and compressibility. And we try to understand the basic ideas of equilibrium between phases, and, through computational exercises, of transformation between them.

The mathematical formalism of statistical mechanics draws both on the laws of mechanics — classical or quantum, as appropriate — and on probability theory and statistics. Bridging the gap between the mathematics and the physics can be difficult, but it is rewarding, for starting from little one can get a lot.

The computer has become an indispensable complement to mathematics in the physical description of many-particle systems. Computational statistical mechanics is not taught in class by the instructor, but the graduate assistant help students to learn it. It forms an integral part of the course that every homework assignment contains a computational problem, and the term project are computational.

Pre-requisites: Thermal Physics & Mathematical Physics I (Mathematical and Computational Toolkit).

Elective Courses

Level 3000

PHY3520 Quantum Mechanics 2

This course is a follow-up on Quantum Mechanics 1, and thus knowledge of quantum mechanics at that level will be assumed. The topics expected to be covered are: (i) identical particles, (ii) angular momentum, (iii) approximation methods, including the variational method, the WKB method, and time-independent and time-dependent perturbation theory, and (iv) scattering theory. If time permits, an introduction to relativistic quantum mechanics will be provided. Students are expected to be comfortable with partial differential equations and linear algebra; knowledge of computational methods at the level of Computational Physics would help but is not mandatory.

Pre-requisites: Quantum Mechanics I

PHY3010 Lab 4: Quantum Mechanics & Statistical Mechanics

Designed to accompany the two theory courses - Quantum Mechanics and Statistical Mechanics, this laboratory course explores their experimental foundations. Students are introduced to some of the experiments which required Quantum Mechanics in order to be explained, notably discrete atomic spectra and the Zeeman Effect. The lab also contains experiments on Brownian Motion and the determination of other properties of matter like magnetic susceptibility.

Pre-requisites: Lab 1: Introduction with Physics through experiment

PHY3110 Mathematical Physics 3

This mathematical physics course develops the use of complex analysis in physics. It develops the subject from an application point of view, and discuss its applications in Fourier transforms, Laplace transforms, Green's functions, differential equations, special functions, etc. It is useful for those wishing to pursue theoretical physics.

Pre-requisites: Mathematical Physics 1 & 2

PHY3150/PHY6150 Computational Physics

This course focuses on developing an algorithmic approach to problem-solving and on translating algorithms to working computer codes. The course starts from the basics of computations and errors, and then discusses both deterministic problems and those involving random numbers, like Monte Carlo.

PHY3850/4850 Introduction to Electronics

Our day-to-day life highly depends on electronic devices. Hence, studying electronics has become more important than ever. In this introductory course on electronics, we aim to study semiconductors and their applications in analog and digital electronics. This course is taught in a hybrid mode, where there are both theory classes as well as six laboratory experiments based on the theories taught in the class.

Level 4000

PHY-6150/4150 Symmetry in Physics: Lie algebras, groups and representations

This graduate level course focuses on the physics of symmetry using the mathematics of group theory and representation theory. Symmetry is a fundamental concept throughout physics, be it particle physics and field theory, condensed matter physics, cosmology, quantum mechanics/quantum information or atomic and molecular physics.

The basic concepts of geometrical symmetry, finite groups and their representations are covered. A physically oriented introduction are given to Lie algebras and Lie groups and their representations. Familiar concepts from quantum mechanics are used repeatedly - for example, the hydrogen atom spectrum and its symmetries arise repeatedly during the study of Lie groups. Rather than develop the general theory of semi-simple compact Lie groups, the important concepts are illustrated through particular examples. For example, the groups $SO(2)$, $SO(3)$, $SU(2)$, $SU(3)$ etc. as they arise in physics and their unique and distinctive features and representations.

Pre-requisites: Quantum Mechanics I

PHY4050/3050 Advanced Lab 5: Experimental Soft Matter Physics / Complex Systems

The aim of this course is to teach you how to set up an experiment from scratch.

- a) You will learn - depending on what experiment you choose - the basic physics of liquid crystals/ transport phenomena in electrolyte systems/ physics of fluid mechanics.
- b) Handling and preparing samples/cells and doing measurements.
- c) Programming some instruments using LabView or Matlab/ basics of image processing etc

The complexity of the course will depend on the level chosen.

All experiments in this course will be related to complex systems (Liquid Crystals, Electrolytes, Non-Newtonian fluids), and the following are the possible experiments:

1. Measuring curvature elasticity in a couple of liquid crystal samples by Optical Fredericks Transition method.
2. Measuring flexo-coefficient by analyzing flexo patterns.
3. Characterizing Convective and dielectric instability in negative and/or small positive dielectric anisotropy liquid crystals.
4. Optical activity of twisted nematic cell.
5. Optical characterization of phase transition in a liquid crystal.
6. Gravity current on a horizontal plane at constant volume and constant flux.
7. IV characterization of electro-convection in an electrolyte cell.

PHY4720/PHY6720 Soft-Matter Physics

Matter exists in four phases-solid, liquid, gas and plasma. However, often the distinction is not so simple and wide ranges of systems seem to fall between the solid and liquid phase! For example, look at our skin; should we consider it as solid, or liquid? The screen of our smartphone and other similar display units; are they solid or liquid? They all fall under the domain of soft matter. In this course of Soft Matter Physics, we study some of the physical principles governing the characteristic behavior of different soft matter systems. This course equips you with the fundamentals of soft matter, and also fulfils your immediate research needs in relevant fields.

PHY4220/3220 The Special Theory of Relativity

The special theory of relativity is among the most beautiful and fascinating topics in physics. Its basic concepts are simple and accessible yet counter-intuitive and deep. It is also a remarkable illustration of that common theme in the study of natural laws - how much follows from a few fundamental physical postulates.

Special Relativity is essential for studying all high energy natural phenomena. Its historic roots lie in Mechanics and Electromagnetism, but it is also the foundation essential for many areas of modern physics including Quantum Field Theory, Particle Physics, Cosmology and Astrophysics. It is the essential steppingstone to understanding the general theory of relativity - Einstein's relativistic theory of gravitation superseding Newtonian gravity.

In this course we start from the very basics and after a detailed study of relativistic mechanics and electrodynamics, end into various related topical themes from quantum mechanics, optics, astrophysics, thermodynamics and hydrodynamics.

Pre-requisites: Classical Mechanics & Electricity & Magnetism in Light of Relativity

PHY6640 Phase Transitions and Critical Phenomena

This course covers the following topics: thermodynamics of phase transitions, different types of phase diagrams, statistical mechanical approach to phase transitions and critical phenomena, mean field theory and Landau theory, scaling and universality, multicritical points, dynamics of phase transitions, quantum phase transitions and nonequilibrium phase transitions.

Pre-requisites: Thermal Physics & Statistical mechanics

PHY-6620/4620 Non-equilibrium Statistical Mechanics

In the absence of any general framework (with no non-equilibrium analogue of entropy), there are several broad approaches to study non-equilibrium systems. This course discusses these general approaches like Langevin equation, Fokker-Planck formalism, Martin-Siggia-Rose field theory, master equation, Boltzmann equation, and hydrodynamics. Some of the well-established results like the fluctuation-dissipation theorem, linear response theory, etc., are also covered. Time permitting, a few specific model systems are discussed, especially to introduce the idea of the renormalization group.

Pre-requisites: Statistical mechanics

Level 6000

PHY6630 An introduction to the Renormalization Group

The course focuses on a general development of the renormalization group (RG) as a systematic way to explore the large scale "emergent" behaviour of different types of systems. Technical aspects of RG are discussed by considering specific cases, like, polymers, quantum systems, and critical phenomena. The approach taken help us to see how RG can lead to a proper description of a system, avoiding any prior knowledge of its phenomenology. We start with the real space RG, and then go over to various avatars of the field theoretic RG. Special topics discussed include, the Kosterlitz--Thouless transition (xy model), critical dynamics, nonequilibrium systems.

PHY6540 Quantum Field Theory 1 – Introduction and Foundations

This course introduces a number of important concepts and foundational topics in the relativistic quantum theory of fields. The focus is on the Hamiltonian formulation of QFT, spin 0 (scalar) and spin 1/2 (spinor) fields, and on simple tree-level processes for the most part.

Description of Astronomy Minor Courses

Core Courses

Level 1000

AST/PHY 1080 Observing the Cosmos

This is a hands-on laboratory course with both indoor and outdoor experiments. Students learn how to handle an optical telescope and a radio telescope, a theodolite, a sextant and other related equipment. They measure the temperature of the Sun and several other bright stars, learn how to measure distances using parallax and how to infer the latitude and longitude of a location from the position of stars. Experiments involving important techniques used regularly in Astronomy, including emission and absorption spectroscopy and the study of Doppler effect are introduced. Astrophotography and quantitative photometry of stars are carried out. The experiments also include the measurement of the hyperfine spectral transition of hydrogen, the most abundant element, using a radio telescope, built in-house.

Level 2000

AST/PHY 2850 The Physics of the Universe

This course introduces the students to the contents of the universe and the physical processes that occur in them. It describes how, starting from the big bang, the universe expanded to its present size, what forces it experienced, the nature of dark matter and dark energy, how galaxies, stars and planets formed, the origin and properties of compact objects such as neutron stars and black holes and the predicted future course of evolution of the universe. The prime role of the General Theory of Relativity in many of these processes is highlighted.

Elective Courses

Level 3000

AST/PHY 3870/4870 Cosmic Messengers- 1 (Theory)

This introduces to the student how cosmic objects can be studied using various messengers, such as electromagnetic waves at a wide variety of bands, gravitational waves and various energetic particles. The course describes how these different signals originate, what information they carry and how to interpret them.

AST/PHY 3880/4880/6880 Cosmic Messengers- 2 (Observation)

This course describes the instrumentation used to detect and analyze astronomical signals and introduces the student to real data acquired with professional astronomical equipment, including space-based observatories. A computation laboratory is used for analyzing the data. This will equip the student to deal with research-level astronomical data streams.

Pre-requisites:

Observing the Cosmos or Physics Lab 1

AST 2840/3840 The Earth and other Planets

In the past decade, the study of planets has taken center stage with the discovery of over 5000 extra-solar planets, which are being examined in detail. This course incorporates the lessons learnt from these studies regarding the formation and evolution of planets including the Earth. The physics of gas giants and rocky planets are introduced, along with their atmospheric properties. Possible conditions for the existence of life is discussed.

AST/PHY 3860 The Violent Universe

Violent phenomena like energetic explosions, jet-like ejection of matter or sudden collapse of stars are very common in the Universe. These phenomena play a key role in controlling the energetics of gas in galaxies and the synthesis and dispersal of heavy elements. Of late, major emphasis has been placed globally on the study of such events and new information is emerging rapidly. This course acquaints the student with the physical processes associated with these violent events and the wealth of information that can be gleaned from them about the evolution of the universe.

Pre-requisites:

Physics of the Universe

PHY-4871/ AST-4871/ PHY-6871 Astrophysical and Geophysical Fluid Dynamics

Fluid dynamics is at the heart of most Astrophysical phenomena, and many of these processes have close parallels in Geophysics. In this course the fluid dynamical principles are introduced with specific applications to Astrophysical and Geophysical problems. The topics covered include hydrostatic equilibrium, accretion flows, shock waves, radiative gas dynamics and magnetohydrodynamics.

Pre-requisites:

Classical Mechanics & Electricity & Magnetism in Light of Relativity

AST/ PHY4830/6830 Computational Astrophysics and Geophysics

Computational modelling is an essential tool in Astrophysics and Geophysics. Comprehension of diverse physical effects including fluid and gas flow under gravity, heat and radiation transport, oscillations and seismology, formation of structures in the universe etc. is being developed through computations. This course aims to introduce the students to some of the basic techniques to perform such computational modelling. At the heart of the mathematical description of these problems are partial differential equations (PDEs). Several common numerical methods used to solve PDEs are introduced, with application examples drawn from Astrophysical and Geophysical settings. Methods of inference from observational data are also discussed, along with techniques of Monte-Carlo simulations.

Pre-requisites:

Familiarity with coding and calculus; basic knowledge of classical mechanics and thermodynamics.

PHY/AST 4810/6810 Introduction to general relativity with application to cosmology

Gravity is different, unlike other fundamental forces its description is not based on “charges” and “fields.” The best description of gravity is provided by the General Theory of Relativity which is a theory of spacetime geometry. This theory not only provides a stunningly successful description of all known gravitational phenomena but is also a theory of immense beauty and sophistication. The main aim of this course is to introduce the fundamental concepts of General Theory of Relativity, a further aim is to apply it to modern cosmology

Pre-requisites:

All core courses of year 1 and 2.

Names of Cross Listed Elective Courses

- ❑ BIO3513/ BIO6313/PHY3513/ PHY6313: Computational and Mathematical Biology
- ❑ BIO3020/PHY3020: Ecology
- ❑ BIO3314/PHY3314: Forces and Motion in Biology
- ❑ ES3901/PHY3901: Remote Sensing
- ❑ CHM-4405/ CHM-6611/ CHM-3405/ PHY-4530 Group Theory and Quantum Mechanics

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